

Ji-Man, Yu

其他

基本信息

姓名: Ji-Man, Yu

所属院校: 韩国科学技术院

地区: South Korea

企业经历: 有

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

纳米电子与先进工艺

低功耗电路与能量管理

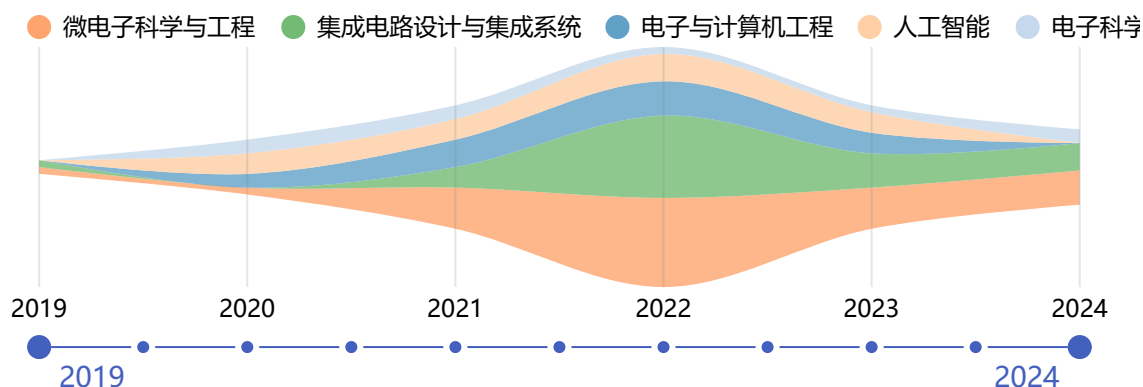
存储器设计与架构

深度学习与神经网络

个人简介

Ji-Man, Yu works at the R&D Organization of SK Hynix. With a verified email at nobelab.kaist.ac.kr, he focuses on CMOS, Neuromorphic Device, and DRAM. His work has been cited 974 times in total and 972 times since 2020, with an h-index of 14 and an i10-index of 22. He has co-authored multiple articles with researchers like Yang - Kyu Choi, Joon - Kyu Han, etc., covering topics such as artificial spiking neuron devices, neuromorphic hardware, and transistor performance improvement.

近年发表的论文



Self-Curable Synaptor With Tri-Node Charge- Trap FinFET for Semi-Supervised Learning

IEEE ELECTRON DEVICE LETTERS

Yu, Ji-Man; Ham, Gyeongdo; Kim, Seong-Yeon; Kim, Jin-Ki; Han, Joon-Kyu; Yun, Seong-Yun; Kim, Seonghak; Lee, Sang-Won; Jeon, Seung-Bae; Kim, Dae-Shik; Choi, Yang-Kyu

被引用次数: 0

2024

Capacitive Synaptor With Overturned Charge Injection for Compute-in-Memory

IEEE ELECTRON DEVICE LETTERS

Kim, Choong-Ki; Phadke, Omkar; Kim, Tae-Hyeon; Kim, Myung-Su; Yu, Ji-Man; Yoo, Min-Soo; Choi, Yang-Kyu; Yu, Shimeng

被引用次数: 1

2024

Bio-Inspired Organic Synaptor with In Situ Ion-Doped Ultrathin Polyelectrolyte Containing Acetylcholine-Like Cation

SMALL

Yu, Ji-Man; Kim, Youson; Lee, Changhyeon; Jeong, Booseok; Kim, Jin-Ki; Han, Joon-Kyu; Yang, Junyeong; Yun, Seong-Yun; Im, Sung Gap; Choi, Yang-Kyu

被引用次数: 0

2024

Capacitive Synaptor with Gate Surrounding Semiconductor Pillar Structure and Overturned Charge Injection for Compute-in-Memory

ADVANCED INTELLIGENT SYSTEMS

Kim, Choong-Ki; Read, James; Shon, Minji; Kim, Tae-Hyeon; Kim, Myung-Su; Yu, Ji-Man; Yoo, Min-Soo; Choi, Yang-Kyu; Yu, Shimeng

被引用次数: 0

2024

Frequency Response to AC Input Signal in a Neuristor

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yun, Seong-Yun; Lee, Sang-Won; Lee, Taegoon; Son, Joon-Ha; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 0

2024

Single Transistor Latch Near 1 V With Asymmetric Biasing in a MOSFET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Sang-Won; Kim, Seung-Il; Yun, Seong-Yun; Han, Joon-Kyu; Yu, Ji-Man; Son, Joon-Ha; Choi, Yang-Kyu

被引用次数: 0

2024

Physically Unclonable Function With a Rough Silicon Channel MOSFET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jung, Dae-Han; Yu, Ji-Man; Ku, Ja-Yun; Yoon, Sung-Su; Kim, Jae-Hun; Han, Joon-Kyu; Kil, Tae-Hyun; Wang, Dong-Hyun; Yeon, Ju-Won; Choi, Yang-Kyu; Park, Jun-Young

被引用次数: 0

2024

Artificial Multisensory Neuron with a Single Transistor for Multimodal Perception through Hybrid Visual and Thermal Sensing

ACS APPLIED MATERIALS & INTERFACES

Han, Joon-Kyu; Yun, Seong-Yun; Yu, Ji-Man; Jeon, Seung-Bae; Choi, Yang-Kyu

被引用次数: 17

2023

Triple-Node FinFET With Non-Ohmic Schottky Junctions for Synaptic Devices

IEEE ELECTRON DEVICE LETTERS

Yu, Ji-Man; Kim, Seong-Yeon; Kim, Jin-Ki; Han, Joon-Kyu; Jeon, Seung-Bae; Choi, Yang-Kyu

被引用次数: 3

2023

Lowering of Schottky Barrier Height in a Vertical Pillar MOSFET by Deuterium Annealing

IEEE ELECTRON DEVICE LETTERS

Yu, Ji-Man; Wang, Dong-Hyun; Han, Joon-Kyu; Yun, Seong-Yun; Park, Jun-Young; Choi, Yang-Kyu

被引用次数: 0

2023

Error reduction of SRAM-based physically unclonable function for chip authentication

INTERNATIONAL JOURNAL OF INFORMATION SECURITY

Kim, Moon-Seok; Kim, Sungho; Yoo, Sang-Kyung; Lee, Bong-Soo; Yu, Ji-Man; Tcho, Il-Woong; Choi, Yang-Kyu

被引用次数: 1

2023

Leaky FinFET for Reservoir Computing with Temporal Signal Processing

ACS APPLIED MATERIALS & INTERFACES

Han, Joon-Kyu; Yun, Seong-Yun; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 6

2023

Improved SOI FinFETs Performance With Low-Temperature Deuterium Annealing

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ku, Ja-Yun; Yu, Ji-Man; Wang, Dong-Hyun; Jung, Dae-Han; Han, Joon-Kyu; Choi, Yang-Kyu; Park, Jun-Young

被引用次数: 4

2023

Self-aware artificial auditory neuron with a triboelectric sensor for spike-based neuromorphic hardware (vol 109, 108322, 2023)

NANO ENERGY

Yun, Seong-Yun; Han, Joon-Kyu; Lee, Sang-Won; Yu, Ji-Man; Jeon, Seung-Bae; Choi, Yang-Kyu

被引用次数: 0

2023

Self-aware artificial auditory neuron with a triboelectric sensor for spike-based neuromorphic hardware

NANO ENERGY

Yun, Seong-Yun; Han, Joon-Kyu; Lee, Sang-Won; Yu, Ji-Man; Jeon, Seung-Bae; Choi, Yang-Kyu

被引用次数: 23

2023

Vertically Integrated CMOS Ternary Logic Device with Low Static Power Consumption and High Packing Density

ACS APPLIED MATERIALS & INTERFACES

Han, Joon-Kyu; Lee, Jung-Woo; Kim, Young Bin; Yun, Seong-Yun; Yu, Ji-Man; Lee, Keon Jae; Choi, Yang-Kyu

被引用次数: 0

2023

3D Neuromorphic Hardware with Single Thin-Film Transistor Synapses Over Single Thin-Body Transistor Neurons by Monolithic Vertical Integration

ADVANCED SCIENCE

Han, Joon-Kyu; Lee, Jung-Woo; Kim, Yeeun; Kim, Young Bin; Yun, Seong-Yun; Lee, Sang-Won; Yu, Ji-Man; Lee, Keon Jae; Myung, Hyun; Choi, Yang-Kyu

被引用次数: 3

2023

Co-Integrated Neuromorphic Devices for Bio-Inspired Compliance Control

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Shin, Hery; Yu, Ji-Man; Han, Joon-Kyu; Choi, Yang-Kyu

被引用次数: 0

2023

Reliability Improvement of Gate-All-Around SONOS Memory by Joule Heat From Gate-Induced Drain Leakage Current

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Jung-Woo; Han, Joon-Kyu; Yu, Ji-Man; Lee, Geon-Beom; Tcho, Il-Woong; Choi, Yang-Kyu

被引用次数: 2

2022

Ultra-fast data sanitization of SRAM by back-biasing to resist a cold boot attack

SCIENTIFIC REPORTS

Han, Seong-Joo; Han, Joon-Kyu; Yun, Gyeong-Jun; Lee, Mun-Woo; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 0

2022

Mnemonic-opto-synaptic transistor for in-sensor vision system

SCIENTIFIC REPORTS

Han, Joon-Kyu; Chung, Young-Woo; Sim, Jaeho; Yu, Ji-Man; Lee, Geon-Beom; Kim, Sang-Hyeon; Choi, Yang-Kyu

被引用次数: 5

2022

An Overturned Charge Injection Synaptic Transistor With a Floating-Gate for Neuromorphic Hardware Computing

IEEE ELECTRON DEVICE LETTERS

Kim, Myung-Su; Kim, Jin-Ki; Yun, Gyeong-Jun; Yu, Ji-Man; Han, Joon-Kyu; Lee, Jung-Woo; Seo, Seokho; Choi, Shinhyun; Choi, Yang-Kyu

被引用次数: 4

2022

Concealable Oscillation-Based Physical Unclonable Function With a Single-Transistor Latch

IEEE ELECTRON DEVICE LETTERS

Jung, Jin-Woo; Han, Joon-Kyu; Yu, Ji-Man; Lee, Mun-Woo; Kim, Moon-Seok; Lee, Geon-Beom; Tcho, Il-Woong; Yun, Seong-Yun; Choi, Yang-Kyu

被引用次数: 3

2022

CMOS Ternary Logic With a Biristor Threshold Switch for Low Static Power Consumption

IEEE ELECTRON DEVICE LETTERS

Han, Joon-Kyu; Yu, Ji-Man; Nam, Seo-Yeon; Choi, Yang-Kyu

被引用次数: 6

2022

Reliability Improvement of Gate-All-Around Junctionless SONOS Memory by Joule Heat From Inherent Nanowire Current

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Jung-Woo; Han, Joon-Kyu; Kim, Myung-Su; Yu, Ji-Man; Jung, Jin-Woo; Yun, Seong-Yun; Choi, Yang-Kyu

被引用次数: 2

2022

A Steep-Slope Phenomenon by Gate Charge Pumping in a MOSFET

IEEE ELECTRON DEVICE LETTERS

Kim, Myung-Su; Yun, Gyeong-Jun; Kim, Wu-Kang; Seo, Myungsoo; Kim, Da-Jin; Yu, Ji-Man; Han, Joon-Kyu; Hur, Jae; Lee, Geon-Beom; Kim, Seong-Yeon; Yun, Dae-Hwan; Choi, Yang-Kyu

被引用次数: 2

2022

Curing of 1-Transistor-DRAM by Joule Heat From Punch-Through Current

IEEE ELECTRON DEVICE LETTERS

Kim, Hyun-Jung; Lee, Geon-Beom; Han, Joon-Kyu; Han, Seong-Joo; Kim, Da-Jin; Yu, Ji-Man; Kim, Myung-Su; Choi, Yang-Kyu

被引用次数: 5

2022

A Junctionless Single Transistor Neuron With Vertically Stacked Multiple Nanowires for Highly Scalable Neuromorphic Hardware

IEEE TRANSACTIONS ON ELECTRON DEVICES

Han, Joon-Kyu; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 8

2022

Thermally Stable Single-Transistor Neuron for Reliable Neuromorphic System

IEEE ELECTRON DEVICE LETTERS

Chung, Young-Woo; Han, Joon-Kyu; Jung, Jin-Woo; Yun, Seong-Yun; Yu, Ji-Man; Lee, Mun-Woo; Choi, Yang-Kyu

被引用次数: 3

2022

Tunable and Reconfigurable Logic Gates With Electrolyte-Gated Transistor Array Co-Integrated With Neuromorphic Synapses

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yu, Ji-Man; Lee, Chungryeol; Han, Joon-Kyu; Lee, Sang-Won; Kim, Moon-Seok; Im, Sung Gap; Choi, Yang-Kyu

被引用次数: 3

2022

Crosstalk reduction in tabletop multiview display with fog screen

ETRI JOURNAL

Jeon, Hosung; Lim, Sungjin; Jung, Minwoo; Yoon, Junghoo; Park, Changhwan; Seok, Junho; Yu, Ji-man; Hahn, Joonku

被引用次数: 1

2022

A Review of Artificial Spiking Neuron Devices for Neural Processing and Sensing

ADVANCED FUNCTIONAL MATERIALS

Han, Joon-Kyu; Yun, Seong-Yun; Lee, Sang-Won; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 91

2022

A Bioinspired Artificial Gustatory Neuron for a Neuromorphic Based Electronic Tongue

NANO LETTERS

Han, Joon-Kyu; Park, Sang-Chan; Yu, Ji-Man; Ahn, Jae-Hyuk; Choi, Yang-Kyu

被引用次数: 30

2022

Artificial Olfactory Neuron for an In-Sensor Neuromorphic Nose

ADVANCED SCIENCE

Han, Joon-Kyu; Kang, Mingu; Jeong, Jaeseok; Cho, Incheol; Yu, Ji-Man; Yoon, Kuk-Jin; Park, Inkyu; Choi, Yang-Kyu

被引用次数: 65

2022

Self-Powered Artificial Mechanoreceptor Based on Triboelectrification for a Neuromorphic Tactile System

ADVANCED SCIENCE

Han, Joon-Kyu; Tcho, Il-Woong; Jeon, Seung-Bae; Yu, Ji-Man; Kim, Weon-Guk; Choi, Yang-Kyu

被引用次数: 0

2022

Synaptic Segmented Transistor with Improved Linearity by Schottky Junctions and Accelerated Speed by Double-Layered Nitride

ACS APPLIED MATERIALS & INTERFACES

Kim, Seong-Yeon; Yu, Ji-Man; Lee, Gi Sung; Yun, Dae-Hwan; Kim, Moon-Seok; Kim, Jin-Ki; Kim, Da-Jin; Lee, Geon-Beom; Kim, Myung-Su; Han, Joon-Kyu; Seo, Myungsoo; Choi, Yang-Kyu

被引用次数: 5

2022

Low-temperature deuterium annealing to improve performance and reliability in a MOSFET

SOLID-STATE ELECTRONICS

Yu, Ji-Man; Wang, Dong-Hyun; Ku, Ja-Yun; Han, Joon-Kyu; Jung, Dae-Han; Park, Jun-Young; Choi, Yang-Kyu

被引用次数: 6

2022

Mimicking biological synaptic plasticity with a leaky charge-trap FinFET

JOURNAL OF MATERIALS CHEMISTRY C

Yu, Ji-Man; Han, Joon-Kyu; Choi, Yang-Kyu

被引用次数: 6

2022

An Artificial Neuron with a Leaky Fin-Shaped Field-Effect Transistor for a Highly Scalable Capacitive Neural Network

ADVANCED INTELLIGENT SYSTEMS

Han, Joon-Kyu; Yu, Ji-Man; Kim, Do-Wan; Choi, Yang-Kyu

被引用次数: 2

2022

A Multiple-State Ion Synaptic Transistor Applicable to Abnormal Car Detection with Transfer Learning

ADVANCED INTELLIGENT SYSTEMS

Yu, Ji-Man; Ham, Gyeongdo; Lee, Chungryeol; Lee, Jae-Hyeok; Han, Joon-Kyu; Kim, Jin-Ki; Jang, Donggon; Kim, Nahyun; Kim, Moon-Seok; Im, Sung Gap; Kim, Dae-Shik; Choi, Yang-Kyu

被引用次数: 3

2022

Investigation of Leaky Characteristic in a Single-Transistor-Based Leaky Integrate-and-Fire Neuron

IEEE TRANSACTIONS ON ELECTRON DEVICES

Han, Joon-Kyu; Kim, Myung-Su; Kim, Seung-Il; Lee, Mun-Woo; Lee, Sang-Won; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 8

2021

3D Stackable Broadband Photoresponsive InGaAs Biristor Neuron for a Neuromorphic Visual System with Near 1 V Operation

2021 IEEE INTERNATIONAL ELECTRON DEVICES MEETING (IEDM)

Han, Joon-Kyu; Sim, Jaeho; Geum, Dae-Myeong; Kim, Seong Kwang; Yu, Ji-Man; Kim, Jongmin; Kim, Sanghyeon; Choi, Yang-Kyu

被引用次数: 22

2021

Ternary logic decoder using independently controlled double-gate Si-NW MOSFETs

SCIENTIFIC REPORTS

Han, Seong-Joo; Han, Joon-kyu; Kim, Myung-Su; Yun, Gyeong-Jun; Yu, Ji-Man; Tcho, Il-Woong; Seo, Myungsoo; Lee, Geon-Beom; Choi, Yang-Kyu

被引用次数: 3

2021

Cointegration of single-transistor neurons and synapses by nanoscale CMOS fabrication for highly scalable neuromorphic hardware

SCIENCE ADVANCES

Han, Joon-Kyu; Oh, Jungyeop; Yun, Gyeong-Jun; Yoo, Dongeun; Kim, Myung-Su; Yu, Ji-Man; Choi, Sung-Yool; Choi, Yang-Kyu

被引用次数: 60

2021

All-Solid-State Ion Synaptic Transistor for Wafer-Scale Integration with Electrolyte of a Nanoscale Thickness

ADVANCED FUNCTIONAL MATERIALS

Yu, Ji-Man; Lee, Chungryeol; Kim, Da-Jin; Park, Hongkeun; Han, Joon-Kyu; Hur, Jae; Kim, Jin-Ki; Kim, Myung-Su; Seo, Myungsoo; Im, Sung Gap; Choi, Yang-Kyu

被引用次数: 49

2021

One Biristor-Two Transistor (1B2T) Neuron With Reduced Output Voltage and Pulsewidth for Energy-Efficient Neuromorphic Hardware

IEEE TRANSACTIONS ON ELECTRON DEVICES

Han, Joon-Kyu; Yun, Gyeong-Jun; Han, Seong-Joo; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 5

2021

Data Sanitization of SRAM by Thermal Distortion

IEEE TRANSACTIONS ON ELECTRON DEVICES

Han, Joon-Kyu; Han, Seong-Joo; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 0

2021

A Temperature Sensor With a Thermillator

IEEE ELECTRON DEVICE LETTERS

Lee, Mun-Woo; Han, Joon-Kyu; Yun, Gyeong-Jun; Yu, Ji-Man; Lee, Geon-Beom; Han, Seong-Joo; Choi, Yang-Kyu

被引用次数: 12

2021

Multi-functional logic circuits composed of ultra-thin electrolyte-gated transistors with wafer-scale integration

JOURNAL OF MATERIALS CHEMISTRY C

Yu, Ji-Man; Lee, Chungryeol; Han, Joon-Kyu; Han, Seong-Joo; Lee, Geon-Beom; Im, Sung Gap; Choi, Yang-Kyu

被引用次数: 5

2021

A Vertical Silicon Nanowire Based Single Transistor Neuron with Excitatory, Inhibitory, and Myelination Functions for Highly Scalable Neuromorphic Hardware

SMALL

Han, Joon-Kyu; Oh, Jungyeop; Yu, Ji-Man; Choi, Sung-Yool; Choi, Yang-Kyu

被引用次数: 8

2021

A Poly-Crystalline Silicon Nanowire Transistor with Independently Controlled Double-Gate for Physically Unclonable Function by Multi-States and Self-Destruction

ADVANCED ELECTRONIC MATERIALS

Yu, Ji-Man; Yun, Gyeong-Jun; Kim, Moon-Seok; Han, Joon-Kyu; Kim, Da-Jin; Choi, Yang-Kyu

被引用次数: 14

2021

A Single Transistor-Based Threshold Switch for a Bio-Inspired Reconfigurable Threshold Logic

ADVANCED ELECTRONIC MATERIALS

Han, Joon-Kyu; Lee, Mun-Woo; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 13

2021

Impact of Post-Metal Annealing With Deuterium or Nitrogen for Curing a Gate Dielectric Using Joule Heat Driven by Punch-Through Current

IEEE ELECTRON DEVICE LETTERS

Park, Jun-Young; Yoo, Tae Jin; Yu, Ji-Man; Lee, Byoung Hun; Choi, Yang-Kyu

被引用次数: 5

2021

Bioinspired Photoresponsive Single Transistor Neuron for a Neuromorphic Visual System

NANO LETTERS

Han, Joon-Kyu; Geum, Dae-Myeong; Lee, Mun-Woo; Yu, Ji-Man; Kim, Seong Kwang; Kim, Sanghyeon; Choi, Yang-Kyu

被引用次数: 54

2020

Quantitative Analysis of High-Pressure Deuterium Annealing Effects on Vertically Stacked Gate-All-Around SONOS Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yu, Ji-Man; Park, Jun-Young; Yoo, Tae Jin; Han, Joon-Kyu; Yun, Dae-Hwan; Lee, Geon-Beom; Hur, Jae; Lee, Byung-Hyun; Kim, Seong-Yeon; Lee, Byoung Hun; Choi, Yang-Kyu

被引用次数: 15

2020

A Study of High-Temperature Effects on an Asymmetrically Doped Vertical Pillar-Type Field-Effect Transistor

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Han, Joon-Kyu; Hur, Jae; Kim, Wu-Kang; Park, Jun-Young; Lee, Seung-Wook; Kim, Seong-Yeon; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 9

2020

Mimicry of Excitatory and Inhibitory Artificial Neuron With Leaky Integrate-and-Fire Function by a Single MOSFET

IEEE ELECTRON DEVICE LETTERS

Han, Joon-Kyu; Seo, Myungsoo; Kim, Wu-Kang; Kim, Moon-Seok; Kim, Seong-Yeon; Kim, Myung-Su; Yun, Gyeong-Jun; Lee, Geon-Beom; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 79

2020

Triboelectric nanogenerator for a repairable transistor with self-powered electro-thermal annealing

NANO ENERGY

Kim, Weon-Guk; Han, Joon-Kyu; Tcho, Il-Woong; Park, Jun-Young; Yu, Ji-Man; Choi, Yang-Kyu

被引用次数: 8

2020

A Single Transistor Neuron With Independently Accessed Double-Gate for Excitatory-Inhibitory Function and Tunable Firing Threshold Voltage

IEEE ELECTRON DEVICE LETTERS

Han, Joon-Kyu; Seo, Myungsoo; Yu, Ji-Man; Suh, Yoon-Je; Choi, Yang-Kyu

被引用次数: 30

2020

学术合作



Myung-Su Kim	合作次数: 12
韩国科学技术院	
Joon-Kyu Han	合作次数: 10
首尔国立大学	
Gyeong-Jun Yun	合作次数: 9
韩国科学技术院	
Sung Gap Im	合作次数: 4
韩国科学技术院	
Il-Woong Tcho	合作次数: 2
韩国科学技术院	

还有 15 个合作者